FD-11 Radiosonde Enclosure

0.6 mm thin-wall SLA housing for a long-duration LoRaWAN picoballoon payload

SOLE MECHANICAL ENGINEER · CAD, PROTOTYPING, ASSEMBLY, TESTING AND PRODUCTION RELEASE

The FD-11 is an ultralight radiosonde developed by Project Picoballoon for long-duration atmospheric observations from high-altitude weather balloons. Every gram of structure is a gram of lift margin, and lift margin sets the ceiling, so the housing is a two-part thin shell printed at **0.6 mm wall thickness** — thin enough that stiffness, printability and clip durability stop being independent problems and have to be solved as one.

6 g	0.6 mm	200	85 × 23 × 18
ENCLOSURE MASS — AGAINST AN 8 G BUDGET	NOMINAL WALL THICKNESS — 0.4 MM AT THE CLIPS AND SNAP-FIT EDGE WALLS	UNITS PRODUCED — EXTERNAL SLA VOLUME MANUFACTURE	ENVELOPE, MM

ARCHITECTURE

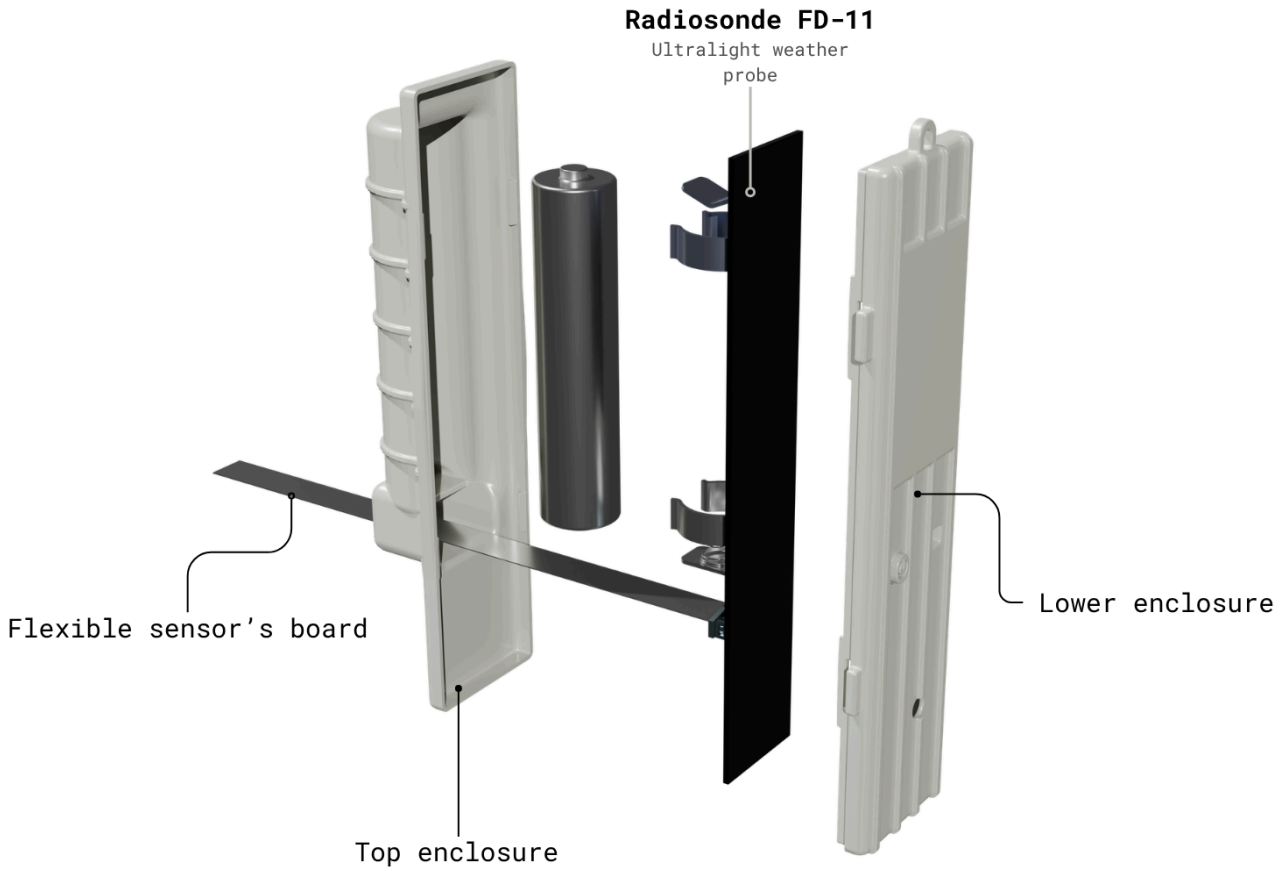


Fig. 1 — Exploded assembly. The shell closes on four internal clips; the AAA cell, the probe and the internal antenna sit inside the 85.3 × 22.6 × 18 mm envelope, and the flexible sensor board leaves the shell through a slot in the parting line, so the sensing elements sit in free air, thermally decoupled from the self-heating electronics.

MASS BUDGET

20 G FLIGHT ASSEMBLY



Enclosure · 6 g
30% OF FLIGHT MASS

Rest of flight assembly · 14 g
70% — LITHIUM AAA CELL, PROBE FD-11, TETHER

The cell was set by the thermal environment and the probe by the electronics design, so the mass budget had to be met by the structure — the only item the mechanical design controls. Individual cell and tether masses are inferred from the 20 g total rather than measured, so the remainder is shown as one segment.

Requirements and the clip problem

REQUIREMENT	RESULT
Mass ≤ 8 g — mission director	6 g
Contain probe, AAA cell and internal antenna — probe design, external	85.3 × 22.6 × 18 mm envelope
Carry the flight train: balloon attaches to the enclosure, probe hangs from it — system architecture	One separation, before the tether-ring fix; none since
Sensors in free air, thermally decoupled from the electronics — measurement requirement	Flexible sensor board exits the shell
Survive multi-day flight, target 50+ days — mission	Longest verified flight: ~41 days

No temperature requirement was formally specified, and there is no dedicated vent — the shell leaks through the clip gaps, the LED opening and the sensor-board slot, so equalization is a consequence of the architecture rather than an intent.

THE CLIP

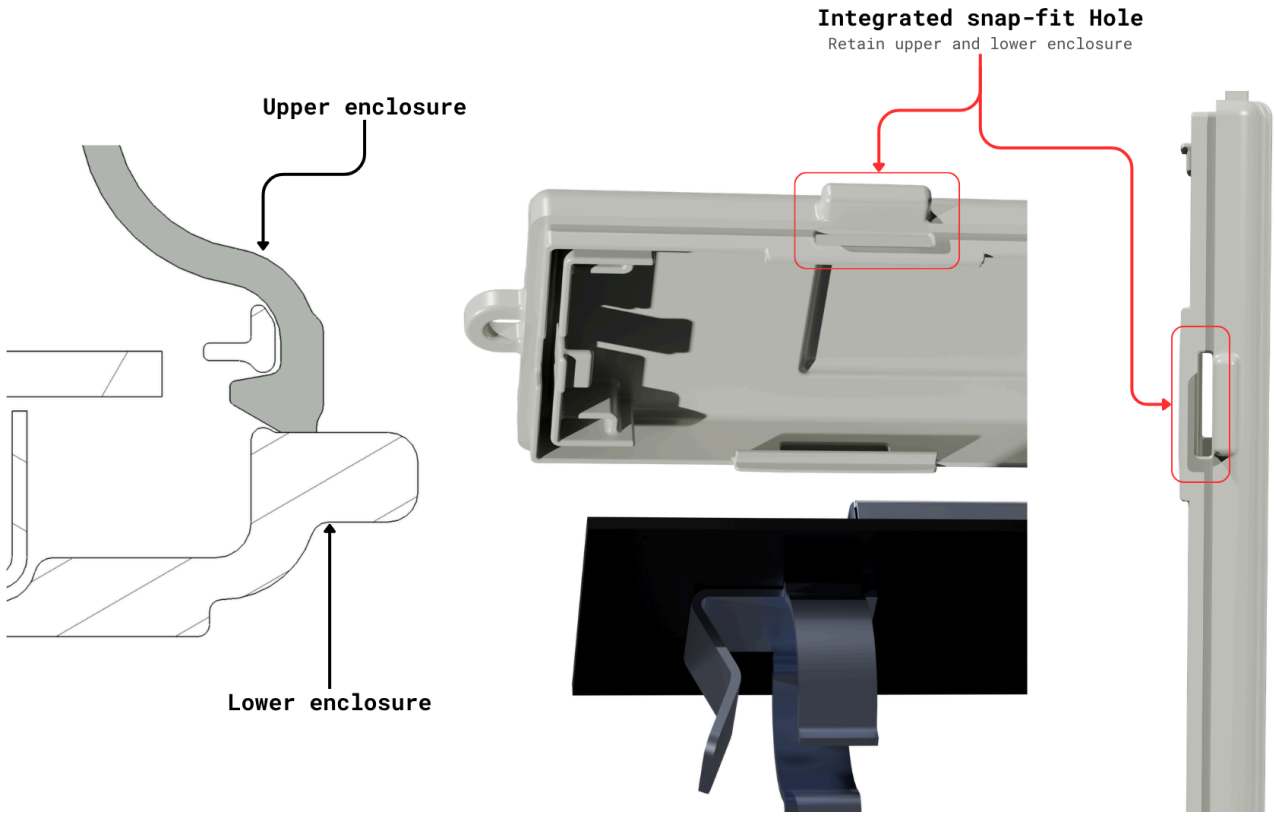


Fig. 2 — Section through the snap-fit. The clip is a 0.4 mm cantilever on the upper shell; it deflects inward as the shells close and returns into the receiving hole in the lower shell. The edge wall around the receiving holes is 0.4 mm, thinner than the 0.6 mm nominal shell, and every opening in it leaves a thin unsupported bridge above it — the ribs restore thickness locally without stiffening the surrounding wall, which would move the crack rather than prevent it.

Seven iterations

3–4 MONTHS

Three iterations were mine: clip geometry, snap-fit dimensions, and local ribs where openings left an unsupported bridge in the 0.4 mm edge wall. Three came from the probe changing underneath the enclosure — the power button, the sensor board and the LED each moved, and each moved an opening. The seventh followed the tether-ring failure in flight.

3 DESIGN · 3 INTERFACE · 1 FLIGHT FAILURE

Clips

Four clips close the shell. They tolerate roughly four or five open/close cycles; the flight condition is closed once and not reopened. No FEA was run. Low prototype cost, short turnaround and uncertainty in thin-wall printed-resin behaviour made empirical iteration the practical development approach.

Print orientation

Prototypes printed at 45°, chosen for printability: a smaller cross-section peels per layer, a closed thin-wall section cannot form a suction pocket, and resin drains. Orientation was fixed by the process, so clip behaviour had to be tuned through geometry.

What flight proved, and what it did not

FAILURE AND FIX

The closed ring where the tether attaches broke once, during flight. The enclosure is the structural link in the flight train, so a ring failure is a loss of the payload rather than a cosmetic problem: the probe separated from the balloon and fell. The ring thickness was increased, and there has been no recurrence on subsequent flights.

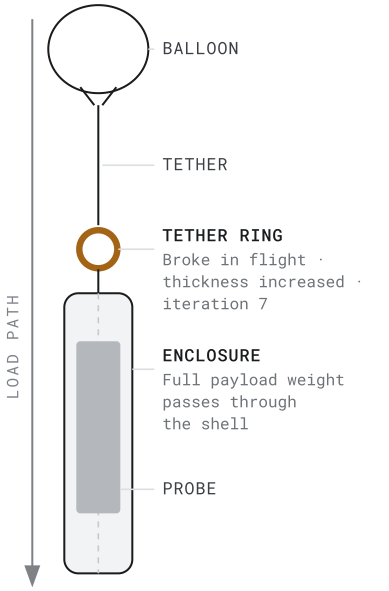


Fig. 3 — Load path. The enclosure is a structural member of the flight train, not a cover.

VALIDATION

1.5 m drop

Onto concrete, fully assembled with cell and boards installed. Pass = shell stays closed and probe still transmitting. Impact 5.4 m/s, ≈0.29 J for the 20 g assembly. Ten units, five drops each.

−78 °C soak

Cold-soaked in dry ice, then visually inspected for cracking. −78 °C is the dry-ice temperature, not a measured enclosure temperature; it is well below the coldest recorded flight temperature. Soak duration: 30 min

15 km

Maximum altitude flown. Minimum recorded temperature: −61 °C

53 flights

Across the program, 53 flights on the frozen design; earlier iterations also flew, including the one where the ring broke. The flights provide operational evidence that the revised ring survived flight loading, but they were not controlled proof-load tests.

The dry-ice soak was a static exposure with visual inspection. It does not cover mechanical loading at low temperature: SLA photopolymers become stiffer and less tough when cold, so clip engagement and impact resistance at flight temperature remain untested.

PRODUCTION

TWO BUILDS, 200 UNITS

Prototyping ran in-house in Formlabs Tough 2K, printed and assembled by hand. Production of 200 units went to JLC3DP in two builds. Internal test launches use CBY resin, sanded only, at €0.56 per set — the flights are not recovered, so cost per flight dominates. Units delivered to clients use 9600 resin with exterior matte white spray paint at €3.28 per set, where surface finish is part of the deliverable.

€0.56 / set

CBY RESIN, SANDED · INTERNAL TEST LAUNCHES · €56 AT 100

€3.28 / set

9600 RESIN, PAINTED · CLIENT DELIVERIES · €328 AT 100

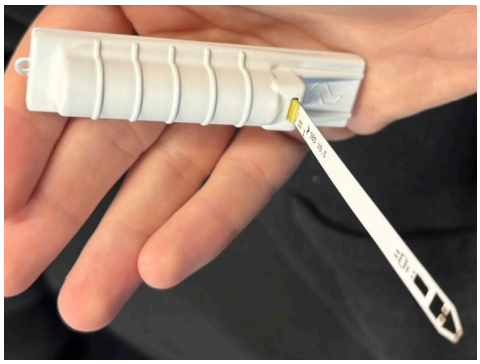


Fig. 4 — Client build: painted upper and lower shells closed on the probe.

The 45° orientation was qualified on the in-house prototypes, but orientation for the production run was chosen by the supplier, so the released parts were not necessarily built the way the prototypes were qualified. Both builds have been flown. Since the resin and finish differ, flight results for one do not directly validate the other.

WHAT I WOULD CHANGE

DESIGN FROZEN

Clip mechanism

The snap-fit is the weakest part of the design. It took the most iterations, caused the early assembly failures, and tolerates only a few open/close cycles. A more robust retention method is the first thing I would change.

50-day survival is unverified

No mission reached that duration. Flights ended when radio coverage was lost, not from hardware failure, so the target remains unproven. Longest verified exposure: **about 41 days**.

UV durability is unverified

The client paint was specified for appearance, not qualified as a UV barrier. Neither build was UV-aged and then tested for clip strength. I would test both after representative exposure to check whether the 0.4 mm clips become brittle.